Fiscal Year:	FY 2004	**Task Last Updated:**	FY 01/23/2005
PI Name:	Fritton, James C Ph.D.		
Project Title:	Bone Adaptation to Precisely Controlled Cyclic Loading		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Physiology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	jcf29@cornell.edu	**Fax:**	FY 212-866-1705
PI Organization Type:	UNIVERSITY	**Phone:**	212-606-1168
Organization Name:	Cornell University		
PI Address 1:	535 E. 70th St.		
PI Address 2:	Research - 5th Floor, Hospital for Special Surgery		
PI Web Page:			
City:	New York	**State:**	NY
Zip Code:	10021	**Congressional District:**	14
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	GSRP
Start Date:	08/01/2001	**End Date:**	07/31/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	van der Meulen, Marjolein (Cornell University)		
Grant/Contract No.:	None		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Because we do not quantitatively understand the relationship between mechanical loading and bone structure, current countermeasures to bone loss are insufficient. Using an in vivo loading model, we propose to determine skeletal adaptation in response to controlled cyclic forces in a weightbearing long bone. Whole bone structural changes will be assessed by microcomputed tomography and histological techniques. The results from these studies will be relevant to understanding and counteracting skeletal adaptation due to space flight, bed rest and paralysis.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	An understanding of how the factors that affect bone adaptation interact is required. Dynamic mechanical forces certainly play a role, as weight-bearing and exercise directly benefit bone. This work will eventually lead to effective countermeasures to the bone loss that occurs during spaceflight, bed rest, paralysis and due to hormonal suppression.
Task Progress:	An experiment aimed at inhibiting hormone related bone loss is being completed. This will be included in the PI's Ph.D. dissertation to be submitted in February 2005.
Bibliography Type:	Description: (Last Updated: 10/29/2019)
Abstracts for Journals and Proceedings	Fritton JC, Doty SB, Myers ER, Wright TM, van der Meulen MCH. "Spatial and temporal mapping of bone mineralization." 50th Annual Meeting of the Orthopaedic Research Society, San Francisco, California, March 7-10, 2004. 50th Annual Meeting of the Orthopaedic Research Society, San Francisco, California, March 7-10, 2004. , Mar-2004
Articles in Peer-reviewed Journals	Fritton JC, Myers ER, Wright TM, van der Meulen MC. "Loading induces site-specific increases in mineral content assessed by microcomputed tomography of the mouse tibia." Bone. 2005 Jun;36(6):1030-8. https://doi.org/10.1016/j.bone.2005.02.013 ; PubMed PMID: 15878316 , Jun-2005